

Zere Bekzhanova

Sustainability & Building Engineer / Computational Designer

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RELOCATING TO AMSTERDAM · OPEN TO THE NETHERLANDS & NORTHERN ITALY

PROFILE

Building engineer across **embodied carbon and life cycle assessment** and **computational design**: parametric modelling in Rhino and Grasshopper, Python automation of BIM data, and cradle-to-gate LCA built on Revit material take-off. MASc Civil Engineering, University of Toronto, first-author paper in **Energy and Buildings** (2026). LEED Green Associate.

EDUCATION

MASc, Civil Engineering · University of Toronto 2023 - 2025

Thesis: embodied carbon and material use across building refurbishment scenarios, using Revit material take-off and cradle-to-gate LCA. Supervisor: Prof. Shoshanna Saxe.

Building & Architectural Engineering · Politecnico di Milano 2022 - 2023

First year completed on a full DSU scholarship, English-medium, before transferring to the MASc.

BEng, Civil Engineering · Nazarbayev University 2017 - 2021

Full scholarship across all years. Two journal papers published as an undergraduate.

EXPERIENCE

Systems & Business Analyst Feb 2026 - Present

ZAAM Construction

- Defined role-level job descriptions and **KPI reporting frameworks across 48 positions**, standardising reporting across project delivery
- Structured the operating system from scratch: **98 process regulations, templates and registers** across six departments, reporting built in SQL, Power BI and Excel

Graduate Research Assistant, Material Use & Embodied Carbon Sep 2023 - Oct 2025

University of Toronto · Centre for the Sustainable Built Environment

- Quantified cradle-to-gate embodied carbon for **five real case-study buildings** (200 to 13,500 m²) across **25 building-phase scenarios**, using Revit material take-off with OpenLCA and ecoinvent
- Built a **Python pipeline** condensing roughly 16,000 raw take-off rows into **4,522 material and assembly line items** for comparative GHG scenario analysis
- Found retained structures kept **60 to 98% of original structural mass**, while embodied GHG still rose **1 to 60%**, driven by new insulation, windows and concrete

Earlier

- Engineering Intern, Schneider Electric** (2021 to 2022). Designed **10 electrical cabinets** to customer specification, from design through to completion.
- Undergraduate Research Assistant, Nazarbayev University** (2019 to 2022). Two peer-reviewed journal papers on recycled concrete aggregate and self-sensing cementitious materials.

SKILLS & CERTIFICATIONS

COMPUTATIONAL	Rhino, Grasshopper (parametric definitions) · Python scripting & automation · Dynamo (basic)
BIM & CAD	Revit · Revit material take-off · AutoCAD
SIMULATION	Climate Studio (environmental) · SAP2000, ETABS (structural, academic)
LCA & CARBON	OpenLCA, One Click LCA · ecoinvent · embodied & whole-life carbon · material flow analysis
DATA	Python (pandas) · SQL · Power BI · Excel · QGIS
CERTIFICATIONS	LEED Green Associate (USGBC) · Construction LCA Specialist (One Click LCA)
LANGUAGES	English (C2, IELTS 8.5) · Kazakh (native) · Russian (native) · Dutch (learning) · Italian (A2) · Turkish (A2)

PUBLICATIONS & ACHIEVEMENTS

- Bekzhanova Z.**, Ruiz-Valero L., Zuluaga S., Saxe S. Material use and embodied GHG in refurbishment and adaptive reuse of existing buildings. *Energy and Buildings* 370, 118075 (2026). First author.
- Two further journal papers (*Nanomaterials* 2021, first author; *Buildings* 2022) and a conference paper at **SBE25, ETH Zurich** (2025). Full list on request.
- Winner, Climate Impacts Hackathon 2024 · European Finalist, Schneider Go Green 2021.**

BEYOND WORK

Tennis and hiking are how I reset, and I have played **classical piano** since childhood. A serious **coffee** habit from my year in Milan, and I follow how **AI and 3D printing** are changing construction workflows. Three countries and two moves for research, so a new city and team are familiar ground.